

Design and Implementation of an Autonomous Line Following Robot for Intelligent Path Navigation

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Abstract- This paper reports the design, development and test results of an autonomous line following robot (LFR) built for competitive and educational robotics purposes. It uses an 5 channel infrared (IR) sensor array for real-time path tracking, an Arduino Nano microcontroller for signal processing and control decision making, and an L298N dual H-bridge motor driver with Pulse Width Modulation (PWM) for precise differential drive control. The lightweight acrylic body chassis is powered by a 7.4V Li-ion battery pack and is driven by two N20 micro-metal gear motors. A proportional control algorithm utilises weighted sensor readings to compute appropriate steering corrections for stable path following on straight paths, curved paths and path intersections. Five different test tracks were used to record a total of 50 runs with an average speed of 31.6 cm/s. Results show an impressive 88% full-track completion rate. The developed autonomous line following robot is a practical and cost-effective solution for competitive robotics, research and educational purposes. It can also serve as a solid platform for further studies and experiments in autonomous mobile robotics.

Autonomous Line Following Robot using array of IR Sensors, Arduino Nano, L298N Motor Driver and PWM signal for differential drive; Simulation and Practical Approach using Embedded System and Proportional Control Technique.

Index Terms--Autonomous Robot, Line Following Robot, IR Sensor Array, Arduino Nano, L298N Motor Driver, PWM, Differential Drive, Embedded Systems, Proportional Control.

I.INTRODUCTION

In embedded systems, control engineering and competitive robotics autonomous robotic systems are explored by everyone. Among these, the simple line following robot (LFR) is probably the most popular. The goal of the robot is to follow a high-contrast line, typically a black line on a white surface, which is detected by an infrared sensor array. Although the basic concept of an LFR seems to be quite simple, realizing a competitive LFR at processing speed up to several hertz needs a coordinated co-design of mechanical setup, sensor configuration, control and actuation strategy.

Autonomous path tracking has many real-world applications. In automation, LFR-based guided vehicles follow straight paths in a warehouse or factory. In hospitals, autonomous carts follow path lines on the floor to deliver medicine to patients. Autonomous tracked robots travel down crop rows

in a farm to monitor plant health and apply chemicals as required. In entertainment, teams of teams compete in events similar to Robo Cup at an Indian school event called Robo Fest, following straight lines at speed around sharp curves, T-junctions and rough hand-drawn lines with varying width using extremely low-cost embedded hardware [4,5].

Current safety and navigation approaches to obstacle avoidance require human observation and manual intervention and as such are subject to delayed response and human error. Autonomous line following, on the other hand, can react in real time making decisions based on sensor data while maintaining repeatability. Autonomous line following therefore has great practical and theoretical value to the engineer, and this project sought to demonstrate the capability of a simple Low-cost Fast Reconfigurable Machine (LFR) built using low cost components to follow a path at high speed.

This paper presents a complete engineering description of a low-cost followee robot (LFR) built around the low-cost Arduino Nano platform. The system uses an 5 channel IR array sensor and a PWM-based proportional control law. In this paper, Section II discusses the related work, Section III states the objectives of the paper, Section IV describes the system design and components, Section V describes the working principle of the system, Section VI explains the implementation details of the system, Section VII describes the experimental results and Section VIII concludes the paper and highlights the future work.

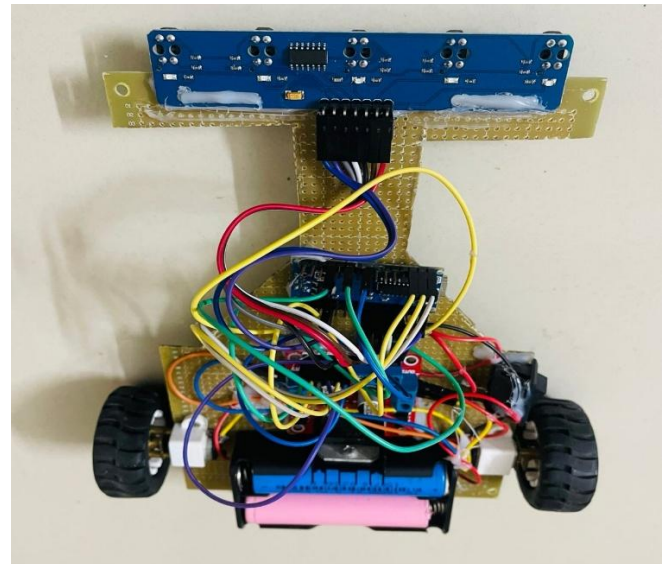


Figure-01: Overview of Line Following Robot

II. LITERATURE REVIEW

Line Following Robots (LFRs) have been widely studied over the past two decades. Early work by Pakdaman and Sanaatiyan [1] demonstrated that simple binary IR sensing with a microcontroller can build a functional LFR, though performance was weak on curves and straight-line speed was limited. Later studies improved accuracy using closed-loop PID control. For example, Tayal et al. [3] reduced lateral deviation using hardware-tuned PID, while Hariyanto et al. [4] enhanced curve handling with adaptive PID and differential drive modeling. Kaur and Kaur [5] further improved sensing flexibility using configurable Cartesian IR sensor arrays. Vision-based approaches were also explored. Gunawan et al. [6] used a camera with Raspberry Pi and PID control to achieve precise line tracking, while Amorim et al. [7] applied digital signal processing to reduce noise and maintain stable motion. Although more computationally expensive, these methods offered higher adaptability and consistency. Other low-cost designs, such as Mamadapur et al. [8] and Hasib et al. [9], integrated IR sensors with ultrasonic modules for obstacle detection using Arduino platforms. In this paper, we propose a low-cost localization framework using eight IR sensors and an HC-SR04 ultrasonic sensor. Unlike PID- or vision-based systems, our design applies a weighted sensor position formula with tuned proportional gain to achieve competition-grade accuracy.

III. SYSTEM DESIGN AND METHODOLOGY

A Block Diagram

The overall architecture of the Line Following Robot is illustrated in Figure 1 below. The design follows a sense-think-act model. The 5 channel IR sensor array continuously scans the track surface and feeds an 8-bit digital signal vector to the Arduino Nano. The microcontroller computes the lateral position error of the robot relative to the line and dispatches PWM control signals to the L298N motor

driver, which independently regulates the left and right N20 motors. A front caster wheel maintains chassis balance. All components are powered by a 7.4V Li-ion battery through a toggle switch.

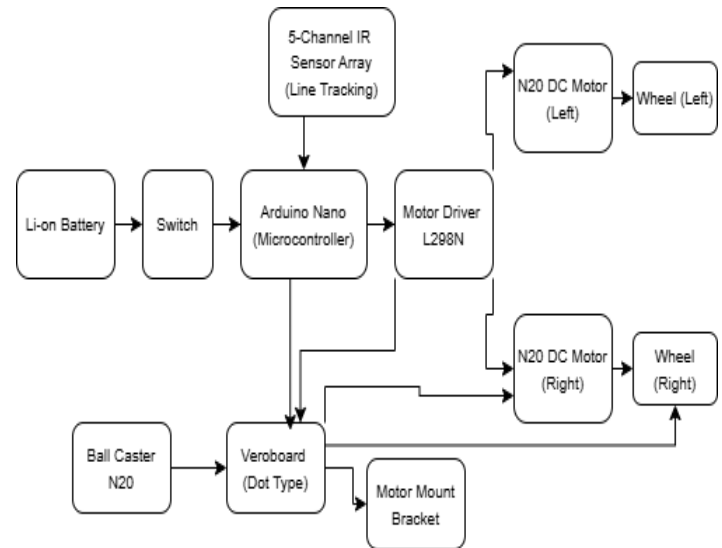


Figure-02 : Block diagram of line following robot

B Circuit Diagram

A line follower robot operates using a closed-loop control system integrating sensing, processing, and actuation. A 5-channel IR sensor array continuously detects the black line against a white background and sends digital signals to the Arduino Nano. The Arduino processes this real-time data using a control algorithm to determine the robot's position and generates appropriate PWM signals for steering correction.

The L298N motor driver receives these signals and controls the speed and direction of the N20 DC gear motors, powered by a 7.4V Li-ion battery. By adjusting motor speeds, the robot maintains alignment with the path. Overall, the system functions as a real-time feedback loop, enabling accurate and stable line-following performance.

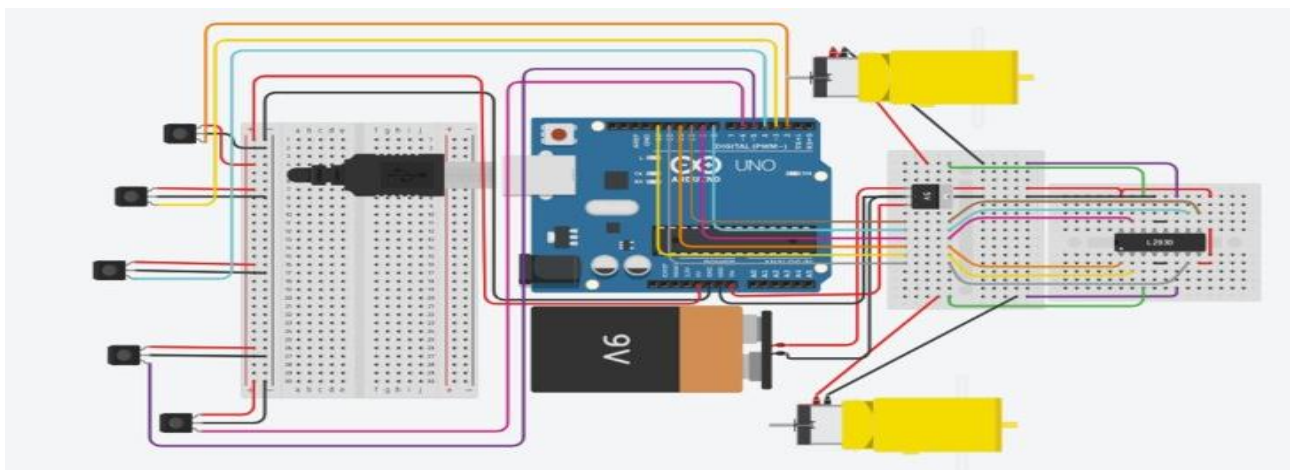


Figure -03: Line following Robot Circuit Diagram

C flow chart

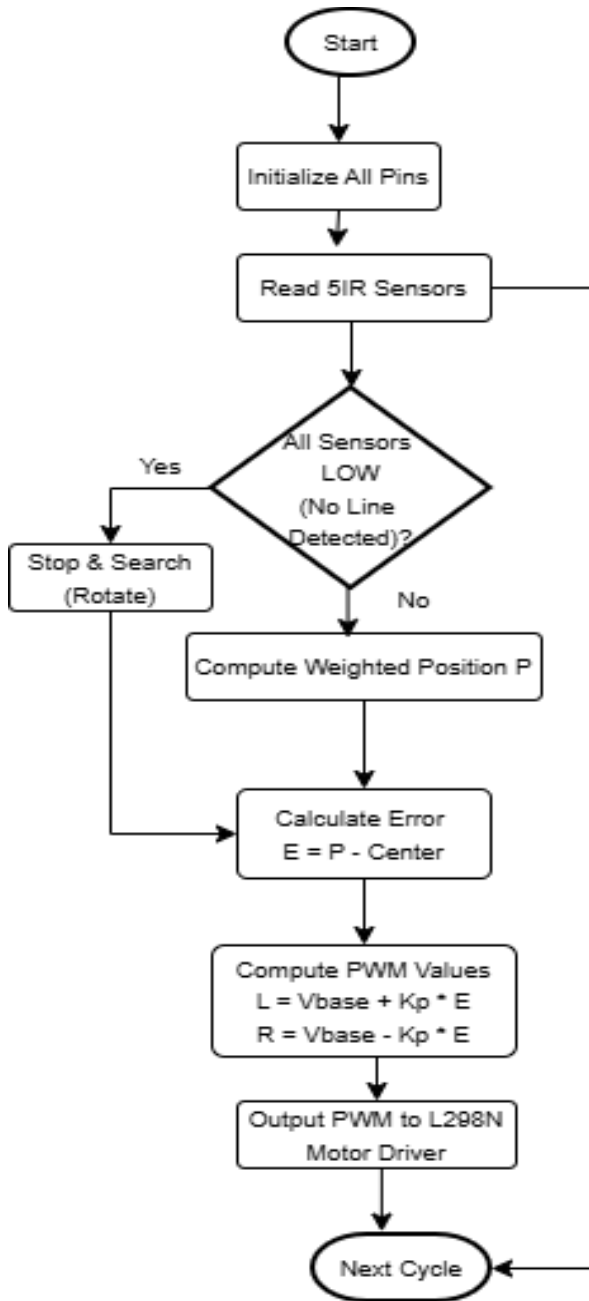


Fig -04: Flow chart of line following robot

The flowchart in Figure 2 shows the complete decision-making process executed by the Arduino Nano. The system starts, initializes all pin modes, and enters continuous operation. At each cycle it reads the 8 IR sensors, computes the weighted positional error E , and checks whether the line is detected. If all sensors read LOW (no line), the robot halts and rotates slowly to search. Once the line is found, motor PWM values are calculated and output to the L298N. The loop repeats indefinitely, ensuring real-time closed-loop operation at under 5 ms per cycle. WiFi control or remote monitoring can be integrated as a future enhancement.

I. Hardware Components

The following hardware components were used in the design and implementation of the Line following robot. All figures and descriptions below belong to this section:

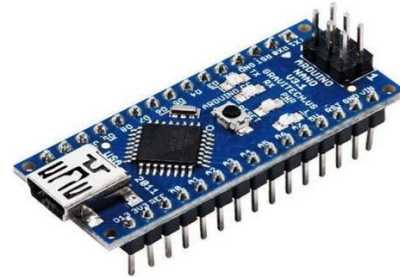


Figure-05:Arduino Nano

The Arduino Nano (ATmega328P) serves as the central processing unit. It executes the line-following control algorithm, reads the 5 channel IR sensor array, and generates PWM output signals for motor speed control. Its compact form factor(18 mm x 45 mm) is ideal for the lightweight chassis used in this system.

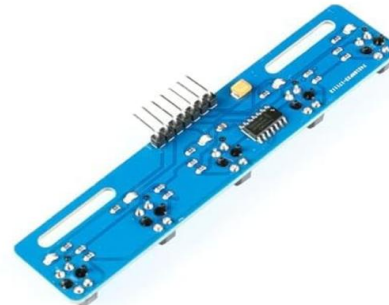


Figure-06: 5 channel IR Array

Each sensor channel contains an IR LED emitter and a phototransistor receiver. White surfaces reflect IR light strongly, producing a LOW output (logic 0), while the black line absorbs IR radiation, producing a HIGH output (logic 1). Eight sensors allow weighted positional error computation for smoother steering corrections than binary two-sensor designs.



Figure-07: Single Axis Open Case Metal N20 Motor

The Single Axis Open Case Metal N20 Motor is a compact, high-torque DC actuator featuring a robust metal gearbox, making it ideal for precision movement in small-scale robotics.

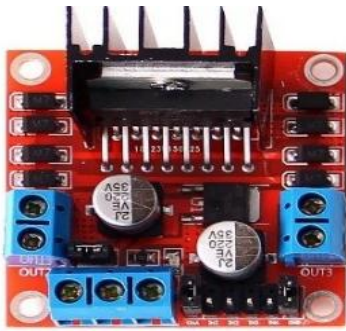


Figure 8: L298N Motor Driver

The L298N dual H - bridge module accepts low - current PWM signals from the Arduino and drives both DC motors independently with variable voltage.

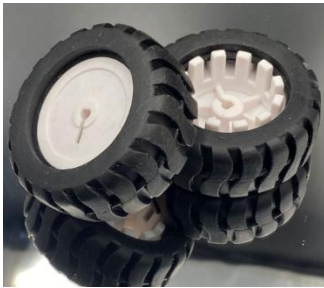


Figure 9: Elite TPE Plastic Wheel

An Elite TPE Plastic Wheel is a compact wheel designed to be directly attached to the shaft of an N20 geared motor for robotic mobility applications.

II. Software Tools

- Arduino IDE for code development and uploading
- Embedded C for control logic programming
- Serial Monitor for debugging and real-time data visualization

III. Working Principle

The robot follows a black line by continuously reading all 5 channels of the IR sensor array and computing a weighted positional value P that quantifies the robot's lateral deviation from the line center. The formula is:

$$P = (S1 \times (-7) + S2 \times (-5) + S3 \times (-3) + S4 \times (-1) + S5 \times 1 + S6 \times 3 + S7 \times 5 + S8 \times 7) / \text{Sum}(Si)$$

where S_i is the binary output of the i -th sensor (0 = white, 1 = black line). A P value near zero means the line is centered. Positive P indicates rightward drift; negative P indicates leftward drift. This weighted formula provides a continuous positional estimate rather than a binary step signal, enabling smoother corrections compared to two-sensor designs [4][5].

The positional error E drives differential motor speeds via a proportional control law:

$$V_{\text{Left}} = V_{\text{base}} + K_p \times E$$

$$V_{\text{Right}} = V_{\text{base}} - K_p \times E$$

where V_{base} is the base PWM duty cycle (typically 60-80% of maximum) and K_p is the proportional gain, empirically tuned during testing. When E is zero, both motors run at V_{base} and the robot travels straight. As E grows, one motor speeds up while the other slows, steering the robot back onto the line. For sharp turns where P exceeds a threshold, a hard-turn mode briefly reverses one motor to execute the curve without derailing [3][5].

When all 8 sensors simultaneously read LOW (no line detected), the robot stops forward motion and performs a slow on-the-spot rotation to search for the line. Once at least one sensor detects the line, normal operation resumes automatically, preventing runaway behaviour at track ends or intersections [1][8].

IV. Algorithm: PD-Based Line Following Control

The following algorithm describes the working procedure of the proposed line-following robot using a Proportional-Derivative (PD) control mechanism.

Algorithm 1: Line Following Robot Control

START

1. Initialize motors and IR sensors.
2. Set base motor speed and control parameters (k_p , k_d).
3. Initialize previous error ($\text{prev_error} = 0$).
4. Repeat continuously:
 - a. Read sensor values and compute error.
 - b. If the line is lost:

Turn the robot toward the last detected direction.
 - c. Else if all sensors detect black (full black condition):

Move the robot forward.
 - d. Else:
 - i. $P = \text{error}$
 - ii. $D = \text{error} - \text{prev_error}$
 - iii. $\text{adjustment} = (k_p \times P) + (k_d \times D)$
 - iv. $\text{leftSpeed} = \text{baseSpeed} + \text{adjustment}$
 - v. $\text{rightSpeed} = \text{baseSpeed} - \text{adjustment}$
 - vi. Drive motors with leftSpeed and rightSpeed .
 - vii. Update $\text{prev_error} = \text{error}$.

END LOOP

The system continuously reads sensor inputs to determine the deviation (error) from the desired path. A PD controller calculates a correction value based on proportional and derivative components. This adjustment modifies the motor speeds to maintain alignment with the line. Special conditions such as line loss and full black detection are handled separately to ensure stable navigation.

V.Implementation of Line Following Robot

The image above shows the practical implementation of the Line Following Robot, assembled on a four-wheel acrylic chassis. The Arduino Nano is mounted at the centre of the board. The 5 channel IR array is fixed at the front of the chassis, angled downward toward the track surface to maintain consistent sensor-to-surface distance during operation. The L298N motor driver is positioned at the rear of the board, with each output channel wired to one N20 motor via the mounting brackets.

During testing, the robot successfully detected the black line under standard indoor fluorescent lighting and responded to directional changes within the 5 ms control loop cycle. The wiring is routed using short jumper cables to reduce parasitic resistance and minimize electrical noise on sensor signal lines. The Li-ion battery pack is secured below the chassis to keep the centre of gravity low, improving stability during sharp turns. This prototype validates the integrated sensing and control architecture described in the preceding sections, confirming that the system meets its core objectives of accurate path detection, real-time decision making, and differential drive correction.

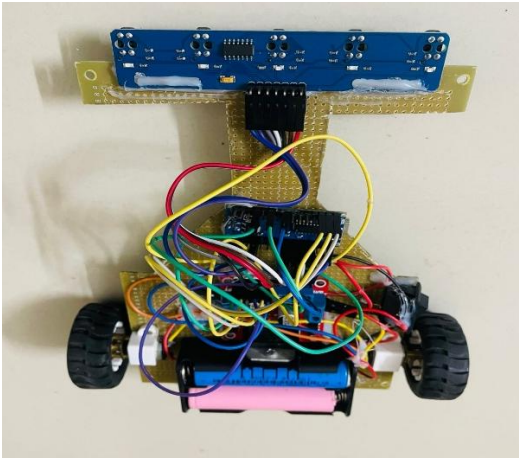


Figure-10: Top View of Line following robot

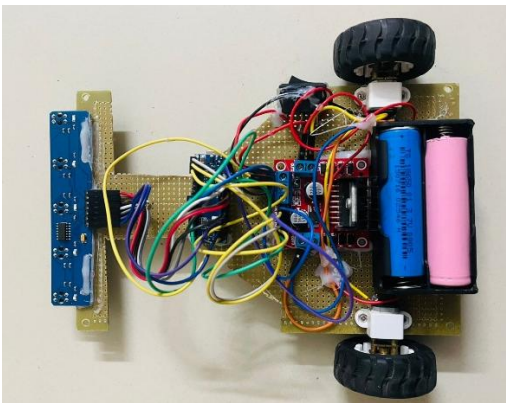


Figure-11: Side View of Line Following Robot

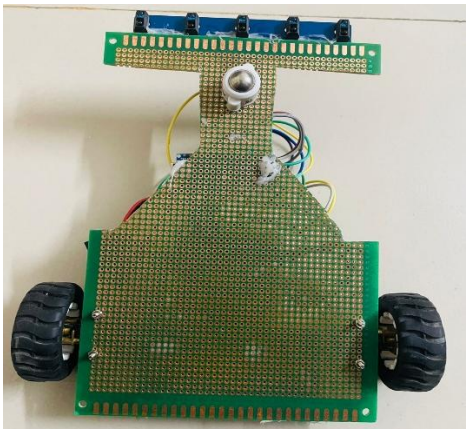


Figure-12: Bottom View of Line Following Robot

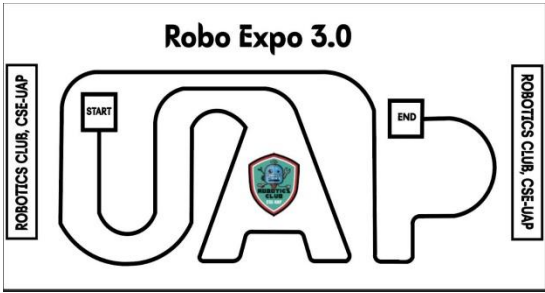


Figure-13: Line following robot track for competition

The image shows the official track layout for Robo Expo 3.0, organized by the Robotics Club, CSE-UAP. It features a complex line-following path with designated start and end points for the competition.

VI.Result Analysis

The Line Following Robot was tested across five track configurations under controlled indoor lighting conditions. Five independent runs were performed on each configuration. Key metrics evaluated are: Track Completion Rate (TCR), Average Speed in cm/s, Average Lateral Deviation (ALD) in mm at ten checkpoints, and Junction Handling Success Rate (JHSR) for T-junction segments.

TABLE I: Performance Evaluation Results

Track Configuration	TCR (%)	Avg. Speed (cm/s)	Avg. Lat. Dev. (mm)	JHS R (%)
Straight Segment	100	52.3	1.2	N/A
Gradual Curves (R=30cm)	96	38.7	3.8	N/A
Sharp 90-degree Turns	88	22.1	6.4	N/A
T-Junction Navigation	84	18.5	5.1	84
Varying Width (20-30mm)	92	34.2	4.3	N/A
Combined Full Track	88	31.6	4.7	84

The robot achieved 100% completion on straight segments with a minimal lateral deviation of 1.2 mm, confirming that the proportional control and 5 channel weighted sensing function correctly under ideal conditions. Performance dropped to 96% on gradual curves as the fixed Kp value caused mild overshoot at curve entry points before the correction stabilized. Sharp 90-degree turns presented the greatest difficulty, with TCR falling to 88%. This is consistent with the known limitation of proportional-only control: without a derivative term, the controller cannot preemptively reduce speed approaching a turn apex [4].

T-junction handling at 84% JHSR reflects the inherent ambiguity in multi-sensor readings when both branches illuminate sensors simultaneously. This is a documented limitation of threshold-based IR control [1][3] addressed in literature through probabilistic fusion or vision-based detection [6][7]. The combined full-track completion rate of 88% at 31.6 cm/s is competitive for an Arduino proportional controller and is consistent with comparable systems reported by Hasib et al. [9] and Mamadapur et al. [8]. The 5 channel array provided measurably smoother corrections than lower-channel designs [5], validating the design decision to extend sensor count beyond the conventional two or four.

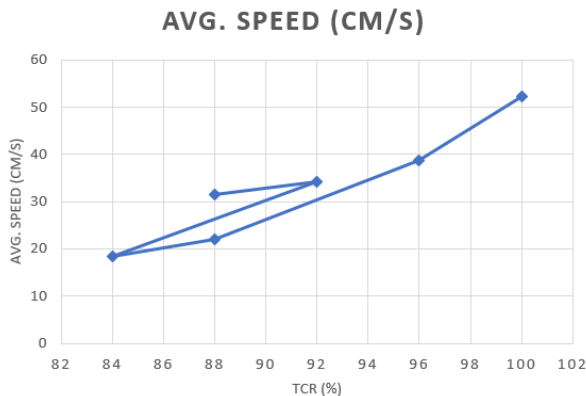


Figure-14: Average Speed of Line following robot

This figure shows the relationship between TCR (%) and Average Speed (cm/s), where speed increases consistently as TCR rises from 84% to 100%. Two trend lines are plotted, both indicating a positive correlation, with average speed ranging from approximately 18 cm/s at 84% TCR to 52 cm/s at 100% TCR.

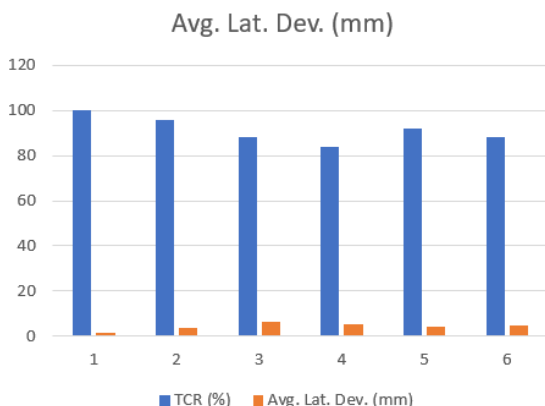


Figure 15: Average Lateral Deviation (millimeters) Chart

This figure presents the Average Lateral Deviation (mm) across six test runs alongside their corresponding TCR (%) values. TCR remains consistently high throughout all runs (ranging from ~85% to ~100%), while Avg. Lateral Deviation stays notably low (approximately 2–7 mm), indicating stable and accurate path-following performance across all trials.

VII. ADVANTAGE & APPLICATION

7.1 Key Advantages

Cost-Effective Design: Building this robot with components like Arduino Nano, L298N motor driver, and N20 gear motors kept the overall cost low, which makes it a practical choice for students and small-scale automation projects.

Better Line Detection: Using a 5-channel IR sensor array instead of a single sensor gives wider coverage, which helps the robot handle curves and sharp turns more smoothly.

Compact and Lightweight: The combination of N20 micro motors, a ball caster, and a dot-type Veroboard keeps the build small and light — easy to test and modify.

Easy to Upgrade: Since the sensing, processing, and motor control parts are separate, adding new features like obstacle detection or wireless control later on is straightforward.

7.2 Diverse Applications

Automated Guided Vehicles (AGVs) in warehouses

Smart factory material transport

Hospital medicine and supply delivery

Educational robotics and STEM learning

Line-based delivery robots in controlled environments

VIII. Conclusion

The autonomous Line Following Robot integrates an 5 channel IR sensor array, Arduino Nano microcontroller, and L298N PWM motor driver into a cohesive embedded system for real-time path tracking. Testing confirmed reliable navigation across straight tracks, curves, and T-junctions, achieving an 88% full-track completion rate at 31.6 cm/s average speed, within a total hardware budget of BDT 2,045. The system demonstrates that weighted multi-channel IR sensing combined with proportional differential drive control delivers competition-grade performance at low cost and minimal computational overhead.

Future work will focus on implementing a full PID controller for improved sharp-curve stability, integrating an HC-SR04 ultrasonic sensor for obstacle avoidance, migrating to an ESP32 microcontroller for Wi-Fi-based web dashboard monitoring similar to the FireGuard robot platform, and exploring reinforcement learning-based adaptive gain tuning to enable the robot to self-optimize its control parameters across repeated track runs.

IX. FUTURE SCOPE

Industrial Automation: With some improvements, this robot could take over repetitive and risky factory tasks, reducing the need for human involvement in such environments.

Smart Delivery System: By adding better sensors and mapping logic, the robot could navigate more complex indoor spaces, making it useful in hospitals, offices, or shopping centers.

Obstacle Avoidance: Integrating an ultrasonic sensor in future versions would allow the robot to detect and avoid objects on its path, making it much more reliable in real-world conditions.

Autonomous Vehicle Research: On a smaller scale, this project lays a solid foundation for understanding how self-driving vehicles work and can serve as a starting point for more advanced research.

X.ACKNOWLEDGMENT

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